

Question Paper Preview

Notations :

- 1.Options shown in **green** color and with  icon are correct.
- 2.Options shown in **red** color and with  icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

AI

Section Id :	640653124671
Number of Questions :	10
Section Marks :	25
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531862691 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536034570.  YES

6406536034571.  NO

Question Number : 2 Question Id : 6406531862692 Question Type : MSQ

Correct Marks : 0 Max. Selectable Options : 0

Question Label : Multiple Select Question

**ASK FOR PRINTED
GRAPH SHEETS**
8 PAGES TWO-SIDE PRINT

Options :

6406536034572. ✓ Printed graph sheets were provided on time.

6406536034573. ✗ Printed graph sheets were provided late.

6406536034574. ✗ Printed graph sheets were not provided.

6406536034575. ✓ I used the graph sheets.

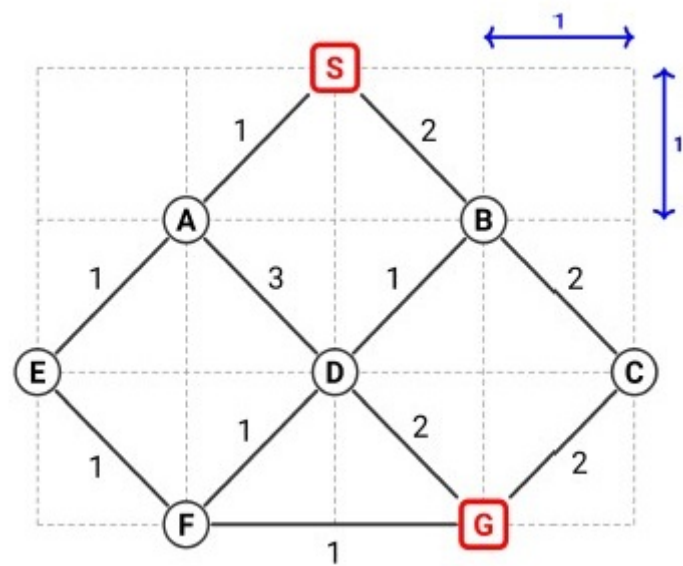
6406536034576. ✗ I did not use graph sheets.

Question Id : 6406531862693 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (3 to 7)

Question Label : Comprehension

SEARCH

The figure shows a map on a uniform grid where each tile is 1x1 in size.
The start node is S and the goal node is G.
The MoveGen function returns nodes in **alphabetical** order.
Use Manhattan Distance as the heuristic function.
Tie-breaker: If several nodes have the same cost, use node labels to break the tie.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 3 Question Id : 6406531862694 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by the Best First Search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z,G

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,B,C,G

S,B,D,G

Question Number : 4 Question Id : 6406531862695 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by A* search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z,G

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,B,D,G

Question Number : 5 Question Id : 6406531862696 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by Branch-and-Bound search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z,G

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,A,E,F,G

Question Number : 6 Question Id : 6406531862697 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For the given map, which algorithm finds the shortest path from S to G?

Options :

6406536034580. ✖ A* Search Algorithm

6406536034581. ✔ Branch-and-Bound Search Algorithm

6406536034582. ✖ Both A* and Branch-and-Bound Search Algorithms

6406536034583. ✖ None of these

Question Number : 7 Question Id : 6406531862698 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Select the correct statement(s) about the given graph.

Options :

6406536034584. ✖ Heuristic is admissible.

6406536034585. ✔ Heuristic is not admissible.

6406536034586. ✖ Heuristic is admissible in some cases and not admissible in other cases.

6406536034587. ✖ There is not enough information to determine admissibility.

Question Id : 6406531862699 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (8 to 9)

Question Label : Comprehension

GAMES: ALPHA-BETA

Consider a game tree with the root node as MAX.

Alpha-Beta algorithm is in mid-flight currently processing a path from the root to a node labeled N.

The snapshot of evals of nodes along the current path is: 5, 17, 7, 11, 9, N.

Where evals are restricted to ODD NUMBERS greater than zero and less than 15.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 8 Question Id : 6406531862700 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Determine the eval of node N that will induce a cutoff to prune its siblings.

Enter an odd number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 17

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

11

13

Question Number : 9 Question Id : 6406531862701 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What type of cut-off is induced by the eval selected for N?

Options :

6406536034589. ✖ Alpha Cutoff

6406536034590. ✔ Beta Cutoff

6406536034591. ✖ Alpha-Beta Cutoff

6406536034592. ✖ None of these

Question Id : 6406531862702 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (10 to 11)

Question Label : Comprehension

GAMES: SSS STAR

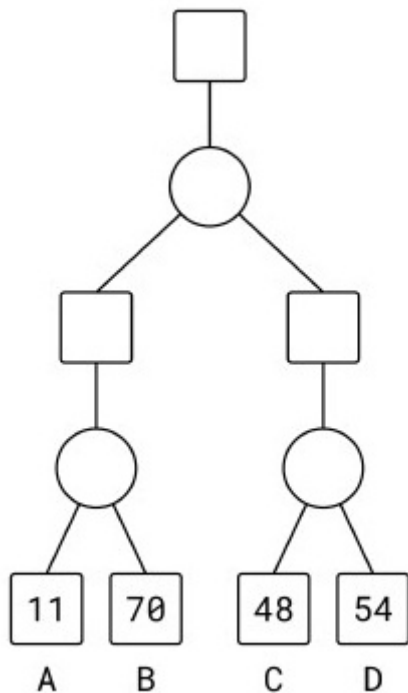
The figure shows a game tree with evaluation function values at the horizon nodes.

The horizon nodes are labeled from A to D.

Use these labels to enter a horizon node or a list of horizon nodes in short answers.

Tie-breaker:

When several nodes carry the same best cost then select the deepest node, if tie persists then select the leftmost of the deepest nodes to break the tie.



Run SSS* algorithm on the game tree, then answer the sub-questions.

Sub questions

Question Number : 10 Question Id : 6406531862703 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes assigned SOLVED status by SSS* algorithm, then enter the labels of those nodes in the textbox, or enter NIL if no nodes are assigned SOLVED status.

Enter node labels as a comma separated list in **alphabetical** order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,B,C,D

Question Number : 11 Question Id : 6406531862704 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes pruned by SSS* algorithm, i.e., horizon nodes removed from the queue by MAX-ancestor.

Enter the labels of those nodes in the textbox, or enter NIL if no nodes are pruned.

Enter node labels as a comma separated list in **alphabetical** order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

NIL

Question Id : 6406531862705 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (12 to 14)

Question Label : Comprehension

PROBLEM DECOMPOSITION

The figure shows an AND-OR decomposition of problem S into smaller problems. The nodes are uniquely identified by labels (S, A, B, C, ...).

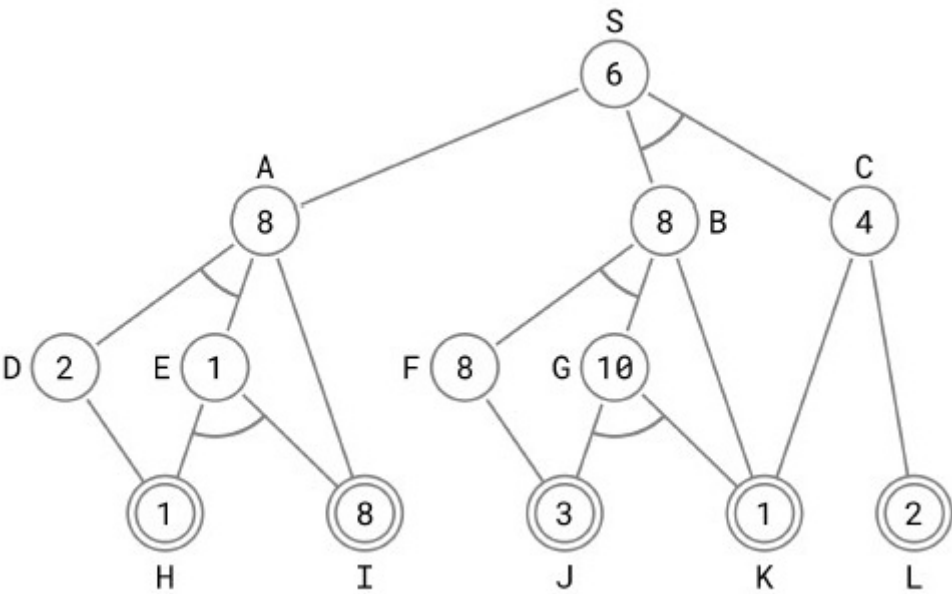
Each node shows the heuristic estimate of the cost of solving that node. Nodes shown in double lines are primitive nodes and their values are actual costs.

A primitive node is added to the graph, with SOLVED status, when its parent is expanded. And therefore, a primitive node is never expanded.

The cost of each edge is 2 units.

Tie-breaker 1: If several nodes have the same cost then break the tie using node labels.

Tie-breaker 2: For AND nodes, select the unsolved branch with the highest cost.



Use AO* algorithm to solve S, then answer the sub-questions.

Sub questions

Question Number : 12 Question Id : 6406531862706 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the nodes expanded by AO* algorithm. List the nodes in the order they are expanded. Observe that primitive nodes are not expanded.

Enter a comma separated list of node labels.
NO SPACES, TABS, DOTS, BRACKETS
OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,A,D,E

A,D,E

Question Number : 13 Question Id : 6406531862707 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

For each node expanded by AO* algorithm, determine the value propagated to the start node S.
Enter the values of S as a list.

Enter a comma separated list of numbers.
NO SPACES, TABS, DOTS, BRACKETS
OR EXTRANEIOUS CHARACTERS.

Answer format: 12,42,17

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

10,9,10,12

9,10,12

Question Number : 14 Question Id : 6406531862708 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What can you conclude about the given AND-OR decomposition?

Options :

6406536034597. ✖ The heuristic is admissible.

6406536034598. ✔ The heuristic is inadmissible.

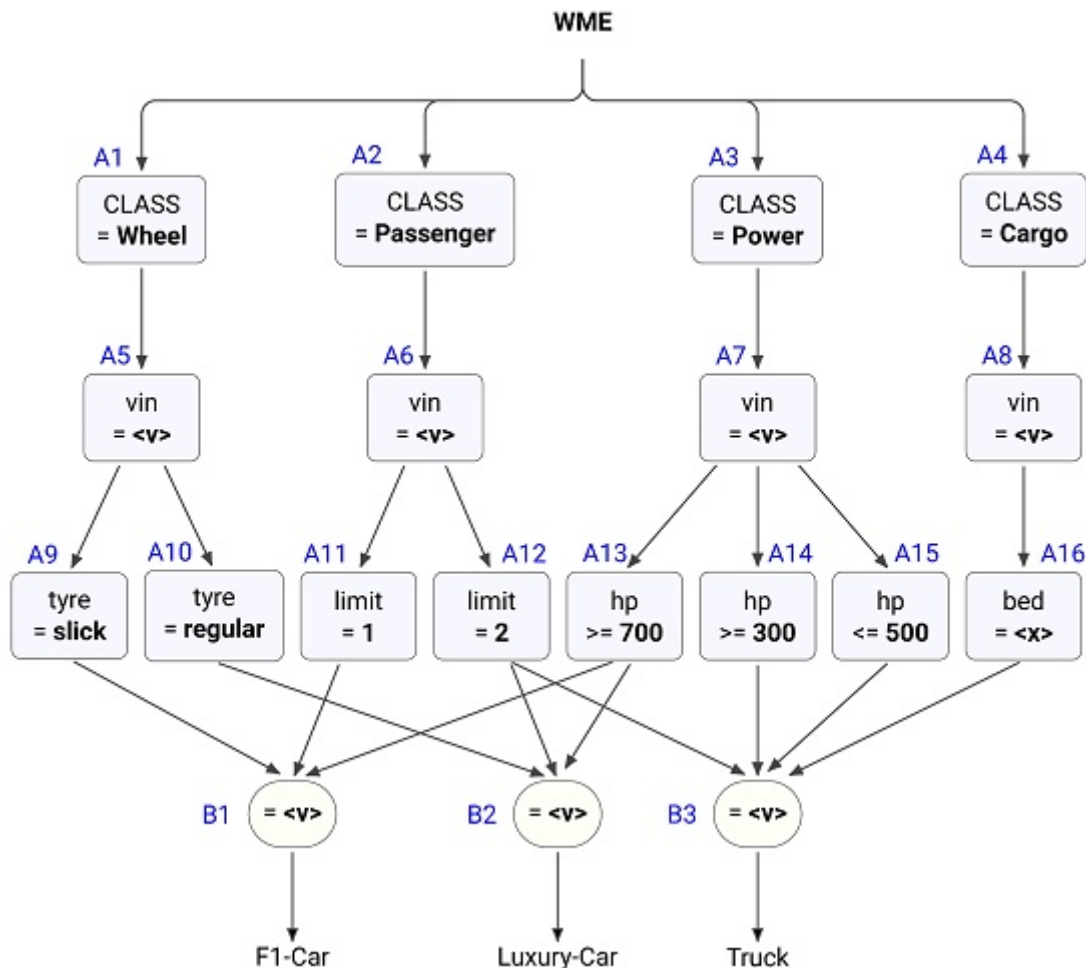
6406536034599. ✖ The heuristic is sometimes admissible and sometimes inadmissible.

Question Id : 6406531862709 Question Type : COMPREHENSION Sub Question Shuffling
 Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
 Question Numbers : (15 to 17)

Question Label : Comprehension

RULE BASED EXPERT SYSTEMS

A Rete Net for classification of vehicles is shown in the figure. Labels A1, A2, A3, ..., A10, A11, A12, A13, ..., and B1, B2, B3 uniquely identify nodes in the network. When required, use the above label ordering to **break ties** and to enter short answers.



The Working Memory contains the following WMEs uniquely identified by timestamps (sequence numbers). Assume that WMEs reside in the appropriate Alpha node, and Beta nodes simply point to WMEs in the Alpha nodes.

101. (Cargo ^vin A1 ^bed flat)
102. (Passenger ^vin A1 ^limit 2)
103. (Passenger ^vin B2 ^limit 2)
104. (Passenger ^vin C3 ^limit 1)
105. (Power ^vin A1 ^hp 400)
106. (Power ^vin B2 ^hp 800)
107. (Power ^vin C3 ^hp 900)
108. (Wheel ^vin B2 ^tyre regular)
109. (Wheel ^vin C3 ^tyre slick)

For each WME identify its location (node label) in the Rete Net, then prepare the conflict set for the first Match-Resolve-Execute cycle, then answer the sub-questions.

Sub questions

Question Number : 15 Question Id : 6406531862710 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following rule-data tuples occur in the conflict set?

Options :

6406536034600. ✓ F1-Car,104,107,109

6406536034601. ✓ Luxury-Car,103,106,108

6406536034602. ✓ Truck,101,102,105

6406536034603. ✗ Truck,101,103,106

Question Number : 16 Question Id : 6406531862711 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Specificity** for conflict resolution then which rule-data tuple will fire in the first round?

Options :

6406536034604. ✗ F1-Car,104,107,109

6406536034605. ✗ Luxury-Car,103,106,108

6406536034606. ✓ Truck,101,102,105

6406536034607. ✗ Truck,101,103,106

Question Number : 17 Question Id : 6406531862712 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Recency** for conflict resolution which rule-data tuple will fire in the first round?

Options :

6406536034608. ✓ F1-Car,104,107,109

6406536034609. ✗ Luxury-Car,103,106,108

6406536034610. ✗ Truck,101,102,105

6406536034611. ✗ Truck,101,103,106

Question Id : 6406531862713 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (18 to 20)

Question Label : Comprehension

AUTOMATED PLANNING 1

Answer the given subquestions.

Sub questions

Question Number : 18 Question Id : 6406531862714 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Consider actions a and b and the two orderings (a then b) and (b then a). Which of the following conditions will make only one of the two orderings feasible?

Options :

6406536034612. ✓ P is in pre(a) and also in del-effects(b).

6406536034613. ✗ P is in add-effects(a) and also in del-effects(b).

6406536034614. ✗ None of these.

Question Number : 19 Question Id : 6406531862715 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In planning graphs constructed by GraphPlan, actions a and b in layer n are mutex _____ .

Options :

6406536034615. ✓ if P in pre(a) and Q in pre(b) are mutex

6406536034616. ✓ if P is in pre(a), in del-effects(a), in pre(b) and also in del-effects(b)

6406536034617. ✓ if P is in pre(a) and also in del-effects(b)

6406536034618. ✓ if P is in add-effects(a) and also in del-effects(b)

Question Number : 20 Question Id : 6406531862716 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In planning graphs constructed by GraphPlan, propositions P and Q in layer n are **non** mutex _____ .

Options :

6406536034619. ✗ if every action pair (a,b) with P in add-effects(a) and Q in add-effects(b) in layer n is **non** mutex

6406536034620. ✓ if at least one action pair (a,b) with P in add-effects(a) and Q in add-effects(b) in layer n is **non** mutex

6406536034621. ✗ none of these

Question Id : 6406531862717 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (21 to 24)

Question Label : Comprehension

AUTOMATED PLANNING 2

The domain description of a Blocks World with a single one-armed robot is given below.

PREDICATES

<code>armEmpty</code>	The arm is not holding any block, it is empty.
<code>holding(X)</code>	The arm is holding X.
<code>onTable(X)</code>	X is on the table.
<code>clear(X)</code>	X has nothing above it, it is clear.
<code>on(X,Y)</code>	X is directly placed on Y.

OPERATORS

`Pickup(X)`: pick up X from the table.

Preconditions: { `armEmpty`, `clear(X)`, `onTable(X)` }

Add Effects : { `holding(X)` }

Del Effects : { `armEmpty`, `onTable(X)` }

`Putdown(X)`: place X on the table.

Preconditions: { `holding(X)` }

Add Effects : { `armEmpty`, `onTable(X)` }

Del Effects : { `holding(X)` }

`Unstack(X,Y)`: pick up X that is directly sitting on Y.

Preconditions: { `armEmpty`, `clear(X)`, `on(X,Y)` }

Add Effects : { `clear(Y)`, `holding(X)` }

Del Effects : { `armempty`, `on(X,Y)` }

`Stack(X,Y)`: place X directly on top of Y.

Preconditions: { `holding(X)`, `clear(Y)` }

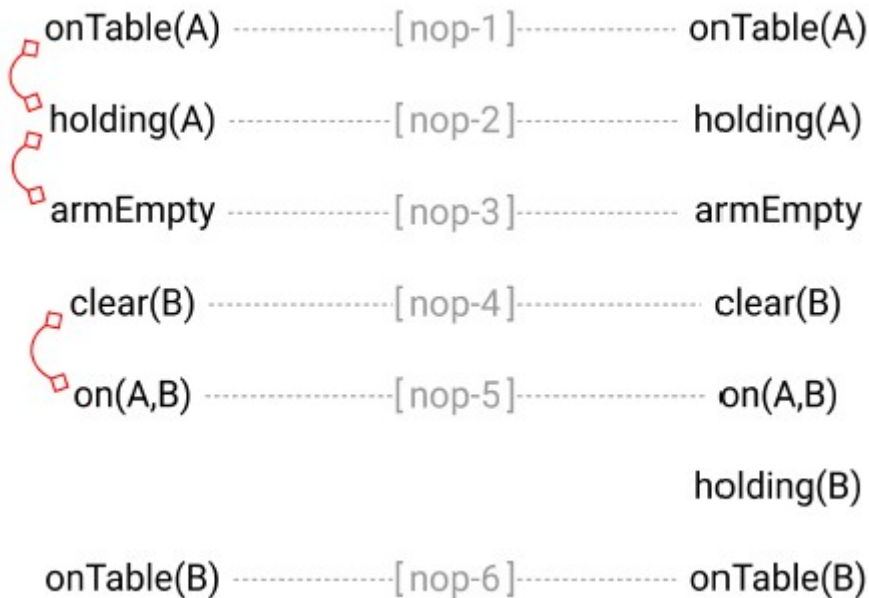
Add Effects : { `armEmpty`, `on(X,Y)` }

Del Effects : { `holding(X)`, `clear(Y)` }

NOTE: variables X and Y must bind to different objects

The GraphPlan algorithm is in mid-flight solving a planning problem, from the planning graph two consecutive propositional layers (layer k and k+1) are presented in the figure.

Both proposition layers are fully populated (no missing propositions).



Mutex proposition pairs in layer k are:
 (onTable(A), holding(A)), (holding(A), armEmpty), (clear(B), on(A,B)).

Populate the action layer and mutex links, then answer the sub-questions.

Sub questions

Question Number : 21 Question Id : 6406531862718 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What can you conclude about layer k?

Options :

- 6406536034622. ✖ Layer k describes the start state of the planning problem.
- 6406536034623. ✔ Layer k cannot describe the start state of the planning problem.
- 6406536034624. ✖ There is insufficient information to comment on layer k.

Question Number : 22 Question Id : 6406531862719 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions in layer k+1?

Options :

- 6406536034625. ✔ Pickup(B)
- 6406536034626. ✔ Putdown(A)
- 6406536034627. ✔ Stack(A,B)
- 6406536034628. ✖ Unstack(A,B)

Question Number : 23 Question Id : 6406531862720 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **mutex** pairs in layer k+1?

Options :

6406536034629. ✓ Putdown(A) and Pickup(B)

6406536034630. ✓ Pickup(B) and Stack(A,B)

6406536034631. ✓ nop-2 and nop-3

Question Number : 24 Question Id : 6406531862721 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

holding(A) and armEmpty in layer k+1 are mutex because _____ .

Options :

6406536034632. ✓ nop-2 and nop-3 are mutex

6406536034633. ✓ nop-2 and PutDown(A) are mutex

6406536034634. ✓ nop-2 and Stack(A,B) are mutex

6406536034635. ✗ nop-2 and Pickup(B) are mutex

Question Id : 6406531862722 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (25 to 27)

Question Label : Comprehension

CONSTRAINT SATISFACTION

Consider a CSP over 3 variables A, B, C (processed in that order) with domains and constraints as shown below.

$$D_A = D_B = D_C = \{1, 2, \dots, 9\}$$

$$R_{AB} = \{ (2, 6), (3, 5) \}$$

$$R_{AC} = \{ (2, 8), (3, 7) \}$$

$$R_{BC} = \{ (5, 7), (5, 8), (6, 7) \}$$

Draw the constraint graph and matching-diagram then answer the sub-questions.

Sub questions

Question Number : 25 Question Id : 6406531862723 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the given CSP arc consistent?

Options :

6406536034636. ✓ Yes

6406536034637. ✗ No

6406536034638. ✗ Cannot be determined

Question Number : 26 Question Id : 6406531862724 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the given CSP is not already arc consistent, make it arc consistent, then check if it is path consistent.

Options :

6406536034639. ✗ It is path consistent.

6406536034640. ✓ It is not path consistent.

6406536034641. ✗ Path consistency does not apply because the constraint graph is cyclic.

Question Number : 27 Question Id : 6406531862725 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Does the given CSP have a solution?

Enter the solution for the variables A,B,C as a comma separated list.
Enter NIL if there is no solution.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 1,2,3

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

3,5,7